

Strain Distribution in OSB and GWB in Wood-Frame Shear Walls

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Abstract: The overall goal of this study is to gain an insight into the load sharing aspect between oriented strand board (OSB) and gypsum wall board (GWB) in shear wall assembly during racking load. Sixteen standard 2,440 × 2,440 mm walls were tested monotonically, in total, out of which 11 (Type A) were sheathed on both sides (OSB on one side and GWB on the other), while five walls were tested without GWB (Type B). Digital image correlation (DIC) was used for data acquisition and analysis which is a full-field, noncontact technique for measurement of displacements and strains. The system returns full-field three-dimensional displacement and strain data measured over the visible specimen surfaces. Overall, these tests suggest that initially during loading of a wall the load is shared between OSB and GWB. However, the proportion of load sharing is not known. As GWB fails first at about 60% of ultimate load capacity of the wall, the load shifts to the OSB panel which resists it until the failure of the wall. The tests also revealed that the load path in wall Types A and B is different and so is the failure pattern. Strains were locally concentrated around the fasteners, while the strains in the field of the panel were below the detection limit of DIC system.

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CE Database subject headings: Shear walls; Wood structures; Frames; Strain distribution.

Introduction

The majority of the buildings built in the United States are wood structures. Traditionally wood structures have performed well during seismic events. The main lateral force resisting system (LFRS) of the wood-frame structure is the shear wall and how well a wood structure performs during an earthquake is largely dependent on the competency of the shear wall. Shear walls are generally made from nominal 2 × 4 or 2 × 6 framing lumber with wood structural panel as sheathing attached to it on one side. It is attached to the framing with the help of dowel type fasteners (nails, screws, staples, etc.) spaced as per strength and stiffness requirements. Sheathing is generally of oriented strand board (OSB) or plywood. Gypsum wall board (GWB) is attached on the other side of the wall for exterior walls. GWB is considered a nonstructural element in LFRS. However, damage assessment after the 1994 Northridge earthquake suggested that most of the shear wall failure was due to cracking and tearing of GWB. Pulling out of nails in OSB and plywood also contributed to failures (Schierle 2002b). The total estimated damage was worth \$40 billion and more than half this amount and 60 fatalities were attributed to the damages in wood-frame structures. Forty eight thousand housing units were rendered uninhabitable (Schierle

2002a). The question that such a huge human and economic loss raised was how to improve existing code provisions and retrofit the existing structures to resist earthquake damages in the future. A better understanding of the behavior of OSB and GWB in a wood-frame shear wall assembly is a logical step in providing for a potential solution.

Propelled by the enormity of damage during the 1994 Northridge earthquake, the consortium of Universities for Research in Earthquake Engineering (CUREE) conducted rigorous testing as a part of an exhaustive study to account for the damages during the Northridge earthquake. One of the findings of this study suggested incorporating the more complicated behavior of finish material effects on shear wall assemblies while considering damage-limitation performance (Cobeen et al. 2004). The contribution of GWB is not included in current design standards (AFPA 2001) but GWB is slightly stiffer than OSB (Table 1) or other sheathing material, but at the same time, it is brittle. Since stiffness attracts load, it is highly probable that a major proportion of the initial load is transferred through the GWB during any seismic event. Being brittle it cracks and subsequently cannot withstand the load after failure. In most modern design this aspect is completely overlooked.

As wood shear walls are the major lateral force resisting system in most buildings, they have been the subject of various studies and research (Filiatrault 2002). However, few studies describe the load sharing between GWB and OSB in a wood-frame shear wall assembly. Similarly, little research on the contribution of GWB to strength and stiffness during a seismic event is available.

Wolfe (1983) tested 30 walls to study the contribution of GWB to the racking resistance of light-frame walls and determined that the contribution can be explained by the law of superposition, i.e., racking resistance of walls with GWB and structural wood panels appeared to equal the sum of contributions of the elements tested independently. Walls tested with panels oriented horizontally were more than 40% stronger and stiffer than those with panels oriented vertically. Finally, Wolfe concluded that GWB could pro-

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Table 1. Modulus of Elasticity of Various Sheathing Materials Obtained from Previous Studies

Material	Modulus of elasticity (GPa)	Source
Plywood	7–3	Younquist (2000)
OSB	5–8	Youngquist (2000)
GWB	5–12	Deng and Furono (2001)

vide significant contribution to the racking resistance when subjected to monotonic loading.

Karacabeyli and Ceccotti (1996) tested 2.44×4.88 m (8 in. $\times$ 16 in.) walls with GWB on one side and OSB on other and concluded that peak load increased but ductility decreased due to the brittle nature of GWB, when compared to only OSB as sheathing. For monotonic tests, they verified the law of superposition proposed by Wolfe (1983) up to a drift of approximately 50 mm. Johnson (1997) concluded that GWB helps resist shear in the low to moderate loading, but plywood resists most of the shear near capacity under monotonic loading. Uang and Gatto (2003) studied the effect of GWB on peak strength, initial stiffness, absorbed energy, and deformation capacity. They observed 12% increase in shear wall strength and 31% decrease in shear wall deformation capacity. Initial stiffness increased by 60% as expected because GWB is stiffer than OSB and attracts more load in the beginning. Toothman (2003) tested 2.44 m $\times$ 1.2 m walls and found similar results as Uang and Gatto (2003) but concluded that the principle of superposition is not valid. While observing failure patterns for the walls sheathed on both sides using nails, GWB panels were always first to fail. This is because of the relative ease with which a nail could tear the sheathing and also because GWB is stiffer than OSB it attracts more load. Toothman concluded that by adding GWB in the structure there is an increase in overall strength, elastic stiffness, and energy dissipation before failure of the wall. He also concluded that GWB provides a substantial amount of shear resistance.

To include GWB in the shear wall assembly design process, it is imperative to investigate the amount or proportion of the load experienced by GWB in a shear wall during a seismic event. A better understanding of the role of GWB in the mechanism of shear wall assembly and the extent of distribution of load during a seismic event are required to increase design efficiency. This study addresses this aspect by testing shear walls under monotonic loading, and hence seeks to analyze the load sharing between OSB and GWB in a wood-frame shear wall assembly.

This project was divided into two parts. The first part addressed the issue of load sharing between OSB and GWB in a wood-frame shear wall assembly. The second part provided insight into differences in performance between walls with and without GWB. Hence these two parts allowed us to investigate load sharing between OSB and GWB. Specifically the objectives of this project were as follows:

1. To evaluate qualitatively the load sharing between OSB and GWB in a wood-frame shear wall assembly;
2. To study the strain profile around fasteners in GWB and OSB;
3. To analyze the failure progression of GWB and OSB; and
4. To study the effects of GWB on shear wall behavior.

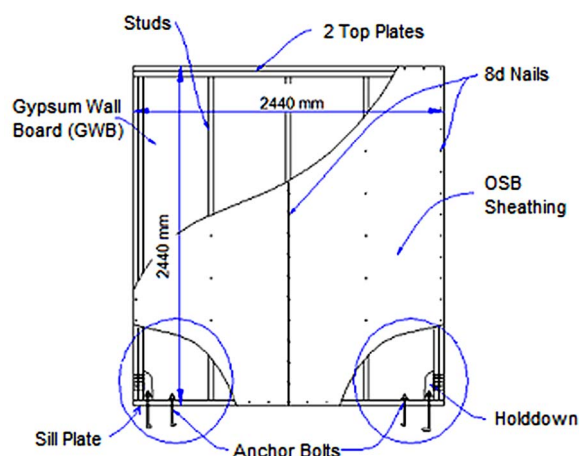


Fig. 1. Schematic of shear wall test specimen

Materials and Methods

Wall Specimens

Shear wall test specimens were designed and constructed in accordance with the 2000 International Residential Code prescribed braced panel construction. All tests were conducted on identical $2,440 \times 2,440$ mm walls constructed using stud grade 38×89 mm kiln dried Douglas-fir framing as shown in Fig. 1. Framing studs were spaced at 610 mm on center, connected to the sill plate and first top plate using two $16d$ (3.33×82.6 mm) nails per connection, driven through the plates and into the end grain of the stud. A second top plate was connected to the first top plate using $16d$ nails at 610 mm on center. The walls were sheathed using two $1,220 \times 2,440 \times 11.1$ mm OSB panels that were attached vertically to the wall frame. The $24/16$ APA rated OSB panels were connected to the wall frame using $8d$ (2.87×63.5 mm) ring shank sheathing nails (Sheather Plus, Stanley) spaced 102 mm on center along the panel edges and 305 mm along the intermediate studs (field nailing). The walls were additionally sheathed with two $1,220 \times 2,440 \times 12.7$ mm GWB panels installed vertically on the face opposite to the OSB structural panels. The GWB panels were attached to the framing with bugle head coarse wallboard screws (2.31×41.3 mm) spaced 305 mm on center along the panel edges and intermediate studs. Sheathing to framing connections was staggered (not shown in Fig. 1) on the end post and top plate. Double end studs were required because walls were anchored with hold-downs, and were connected together using $16d$ (3.33×82.6 mm) framing nails at 305 mm on center. Framing nails were full round head, strip cartridge, and smooth shank SENCO nails that were driven using a SENCO SN 65 pneumatically driven nail gun. Sheathing nails were Stanley Sheather plus nails driven pneumatically as well.

Test Setup

The test setup is shown in Fig. 2. Specimens were bolted to a fabricated steel beam firmly attached to the strong floor to simulate a fixed foundation. Specimens were loaded using a 490 kN (110 kip) servo controlled hydraulic actuator with a 254 mm total stroke, and controlled by an MTS 406 servo controller. The hydraulic actuator was attached to the strong wall and supported by a 102 mm hydraulic cylinder. This allows the actuator to raise and lower freely during the test without creating additional vertical

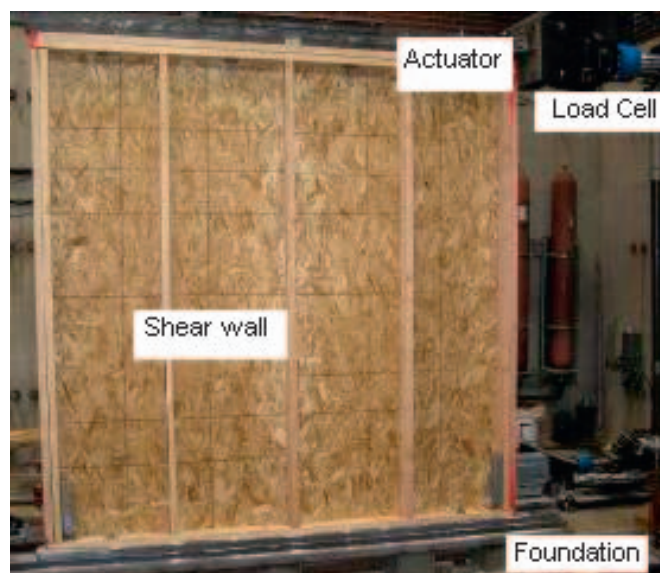


Fig. 2. (Color) Test setup in lab

loading on the wall. A 111.2 kN (25 kip) load cell attached to the piston provided force measurements. A steel C-channel, laterally braced to the strong wall, was attached to the load cell and hydraulic actuator. The C-channel was connected to the top plate of the wall using four evenly spaced 12.7 mm (0.5 in.) A307 bolts installed through both top plate members. To ensure a tight non-slip bolted connection, 12.7 mm (0.5 in.) holes were drilled in the top plates after the walls were positioned.

The data acquisition system connected to the test frame consisted of eight channels of position and load readings. The data from all eight channels were recorded with a computer using a LabView 6i program. Load readings are obtained from a load cell attached to the hydraulic actuator (Channel 1), while deflection at the top of the wall is transferred by the actuator's internal position sensor (Channel 2). The remaining six channels were not used.

Monotonic Testing

Monotonic tests were based on the ASTM E564-00 (ASTM 2000) test protocol. This protocol requires ultimate load to be reached in

no less than 5 min. All walls were tested at 0.76 mm/s. This corresponded to a time to failure of approximately 7 min.

Data Acquisition

An optical measurement instrument based on digital image correlation (DIC) was used to capture and analyze data. DIC is a full-field, noncontact technique for measurement of displacements and strains. The setup consisted of a pair of cameras arranged at an angle to take stereoscopic images of the area of interest as shown in Fig. 3. The cameras were externally triggered and connected to a computer where data were recorded. Image files of undeformed and deformed specimen obtained with the DIC setup were analyzed using proprietary software named Vic 3D (Correlated Solutions Inc., 2005). To calculate displacement at any point, a small subset of pixels was used. This subset has a unique light intensity pattern and the DIC software searches the best matching subset in the image of deformed specimens, using mathematical correlation of intensity patterns, from undeformed specimen images. Once the correlation is finished the system returns full-field three-dimensional (3D) displacement data measured over the visible specimen surfaces and then calculates strains. Surface topography, displacement maps, and strain profiles are obtained from the software. Numerical data for any selected point or area in the image could be extracted from the output files so that some other program (e.g., Excel, etc.) may be used to analyze the data for that area.

Previous research has analyzed displacement fields (Ambu et al. 2005) or crack propagation (Samarasinghe et al. 1996) and others have validated the system by conventional methods or mathematically with a model (Sadeq 2002; Choi and Shah 1997). There is a lack of literature using DIC for larger sample sizes. As of now the application of DIC is limited to small samples with a viewing area of 100×75 mm for concrete samples (Choi and Shah 1997) or 4×5 cm (Samarasinghe and Kulasiri 2000). This study is an attempt to use it for 250×250 mm areas in a $2,440 \times 2,440$ mm shear wall thereby concentrating on strain near fasteners.

Data Analysis

The theory of digital image correlation has been described in detail by several researchers and a detailed treatment of the sub-

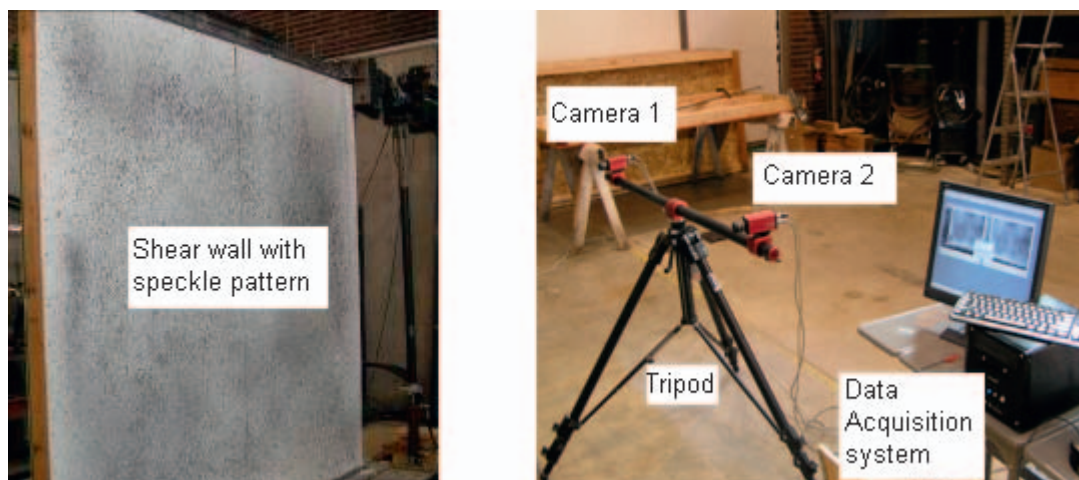


Fig. 3. (Color) DIC setup for shear wall test

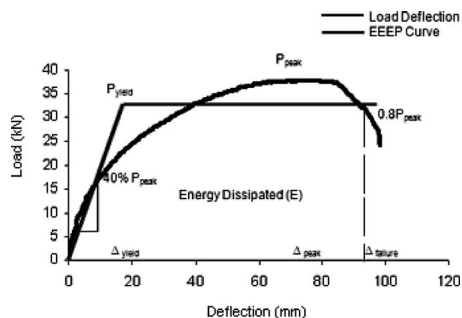


Fig. 4. Equivalent energy elastic-plastic curve

ject can be found in Sutton et al. (1983). The underlying principle of DIC is that the points on the undeformed surface can be tracked to new positions on the image of deformed surface using a least square error minimization technique. It allows measurement of large deformations and strains, far beyond elastic limits of materials. Therefore failure initiating events and at failure strain development may be observed and analyzed. However postfailure, as the specimen undergoes large deformation in a small amount of time, the DIC system finds it hard to track the displacement because they occur between frames acquired by the cameras. Hence postfailure strains, as returned by Vic3d, are large and erratic.

Strains are determined by calculating gradients of displacements, u , v , and w by correlating the position of speckles in a Cartesian coordinate system. Values of various displacement gradients ($\delta u/\delta x$, $\delta v/\delta y$, $\delta w/\delta z$, $\delta u/\delta y$, $\delta v/\delta x$, etc.), which are used to derive strains, are subsequently calculated. Strains are termed as e_{xx} , e_{yy} , e_{xy} , e_1 , and e_2 that correspond to strain in x , y directions, shear strain, and major and minor principle strains, respectively. In this study strains in major principle directions (e_1) are considered as they represent the maximum normal strain at a plane and are dependent on the strain in global x and y directions and also on the shear strains at that point.

An area of interest for numerical data analysis was chosen. The selection of area of interest was based on the magnitude of local principle strain in that area, hence the area which encountered the maximum principle strain on corresponding sides, i.e., on OSB and GWB, was selected as the area of interest to extract after preliminary analysis. The numerical data underlying that area of interest was extracted by Vic3D and then analyzed using other data analysis software (Excel). The area of interest was rectangular in shape, and included 40 data points and had a physical area of 80–100 mm². A detailed description and illustration can be found in Sinha (2007).

Table 2. EEEP Parameters

Parameters	Units	Description
P_{peak}	kN	Measured peak load
Δ_{peak}	mm	Measured displacement at peak load
Δ_{yield}	mm	Calculated yield displacement from EEEP curve (P_{yield}/K_e)
$\Delta_{failure}$	mm	Measured postpeak displacement at 80% peak load
K_e	kN/mm	Calculated elastic shear stiffness ($0.4P_{peak}/\Delta_{0.4P_{peak}}$)
E	J	Calculated energy under the curve to failure
μ		Calculated ductility factor $\Delta_{failure}/\Delta_{yield}$
D		Calculated ductility ratio $\Delta_{peak}/\Delta_{yield}$

Table 3. Test Matrix

Area number	Description of areas	Number of walls	
		Type A	Type B
I	Compression corner	2	1
II	Uplift corner	2	1
III	Center of sill	2	2
IV	Mid wall	2	1 ^a
V	Middle of top plate	1	1 ^a
VI	Intermediate studs	2	0

^aTwo areas were imaged on one wall simultaneously.

EEEE Curve

An analysis of load-deflection curve and equivalent energy elastic-plastic (EEEE) curve provides useful tools to calculate various parameters of the walls. An EEEP curve is a perfectly elastic-plastic representation of the actual response of the specimen. The curve is plotted such that it equals the area under the load-deflection curve until failure. This allows a direct comparison of wall performance on energy basis. Fig. 4 shows the various points of interest used to derive the EEEP curve. The parameters derived from the EEEP curve are listed in Table 2.

Test Matrix

A total of 16 walls were tested monotonically for the project as shown in the test matrix in Table 3. Eleven walls were sheathed on one side with OSB and the other side with GWB (Type A). Five walls were tested with OSB on one side and no sheathing on the other (Type B).

Fig. 5 shows the nailing schedule for OSB and GWB and the areas imaged during the wall tests are marked accordingly. The marked area and corresponding roman numerals are the zones that were imaged. The number of walls tested for each zone is listed in Table 3. Fig. 5 also shows the fastener configuration and numbering scheme on the OSB side. The fasteners on the GWB side are referred to as the same number as that for the OSB side, but preceded by a prefix S (for screws) and are shown as fasteners filled with black.

Results and Discussion

Displacements

Basic data obtained after correlation of images and the displacement in global x , y , and z directions termed as u , v , and w , respectively. Fig. 6 shows a typical load-displacements diagram

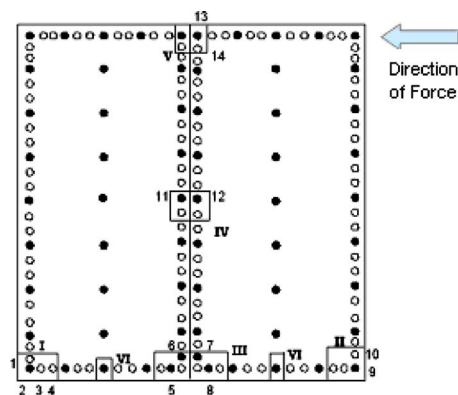


Fig. 5. Nailing schedule for OSB and GWB showing imaged areas

for the uplift corner of GWB in wall Type A. As seen in the graph the vertical displacement (v) is greater than the horizontal displacement (u). Being the uplift corner the vertical displacement here should be more than the horizontal displacement. As seen in Fig. 6 the out of plane displacement (w) is negligible as compared to u and v , which is expected as the panel does not deform much out of plane, hence a low value of w . Displacement graphs for other locations on the wall are given in Sinha (2007) and they all show expected trends.

Load Sharing

OSB and GWB (Wall Type A)

Fig. 7 represents comparative global load versus local strain diagram for the GWB and OSB panels near the uplift corner (Fasteners 9 and S9). As shown in Fig. 7, OSB experiences lower strains than GWB near the uplift corner throughout the entire period of loading indicating that GWB undergoes more deformation than OSB for a given load.

As the shear wall assembly is loaded GWB, which is stiffer than OSB (Table 1), attracts considerable load it undergoes more deformation which results in higher strains until it fails. The start of failure of GWB near connection S9 is observed as scattering of data points circled in Fig. 7 which is clearly observed at around 25 kN. An apparent change of slope is observed in the curve for OSB at 25 kN. The reason for the change in slope is attributed to load shift toward OSB from GWB as GWB around fastener S9 starts to fail. As the load reaches 30 kN the connection on the GWB side (S9) fails, resulting in very high values of optically

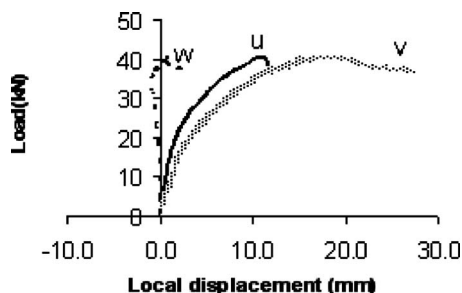


Fig. 6. Load versus local displacement curve for uplift corner in GWB

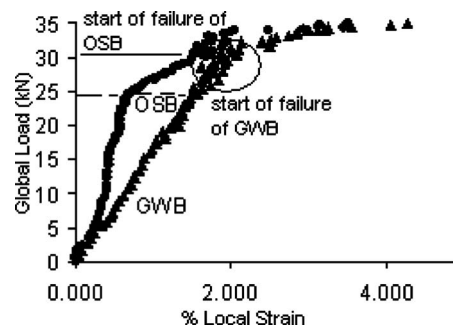


Fig. 7. Global load versus local principle strain for uplift corner (Fasteners S9 and 9)

measured strains. Once the GWB fails, the paper cover tears apart causing the material within to rupture and fall out, resulting in very high values of optically measured strains.

From Fig. 7, consider a ratio of principle strains on corresponding areas in GWB and OSB. Up to 25 kN, which is the linear range of the load-strain curve, the strain near fastener S9 is four times higher than that of the corresponding OSB side Fastener 9, hence the load in GWB is higher than that carried by OSB. Beyond 25 kN, strain in the OSB increases at a faster rate. This continues up to complete failure of GWB at around 30 kN. Therefore it can be concluded that GWB transfers load during initial low loading, but at higher loads OSB transfers most of the load. This is what Johnson (1997) had speculated, that GWB helps resist shear in the low to moderate loading, but plywood (structural sheathing) resists most of the shear near capacity under monotonic loading.

The ratio of strain distribution is different for different regions of the wall because it is based on two factors, namely load path and connection stiffness, of both panels. However for most of the fasteners the strain in GWB is higher than that of OSB for the initial loading period and can be found in Sinha (2007).

OSB (Wall Types A and B)

Figs. 8 and 9 are global load versus local strain plots for corresponding areas in wall Types A and B around Fastener 9 and 5, respectively. The strains in OSB of Type B walls are much higher as compared to Type A walls at any given load, as observed from Figs. 8 and 9. The strain in OSB panels of Type A walls is low for the initial period of loading and at around 25 kN starts increasing at a faster rate until failure of the wall. The onset of strain in OSB of wall Type A is delayed because of the presence of GWB which attracts load initially. In both the graphs (for wall Type A) a

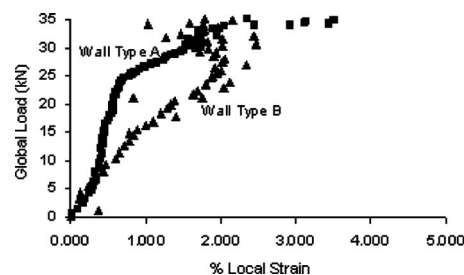


Fig. 8. Global load versus local principle strain (in OSB) plot for Fastener 9

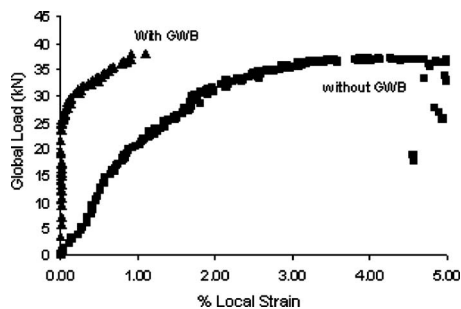


Fig. 9. Global load versus local principle strain in OSB plot for Fastener 5

change of slope is observed at 25 kN which signifies a load shift from GWB to OSB as GWB starts to fail around this load for wall Type A.

As observed previously (Fig. 7) the connection at GWB side in wall Type A starts to fail around 25 kN. At a similar stage of loading a change of slope is observed in the OSB side for Type A walls signifying shifting of load from GWB to OSB as the connections shows signs of failure. Fastener 9 on both types of walls fails early (Figs. 7 and 8), close to 30 kN for GWB in wall Type A and at around 23 kN in OSB for wall Type B, as compared to the rest of the fasteners as after that the high and erratic value of strains are returned by the optical system. The onset of strains was the earliest near Fastener 9 and failure is also initiated in this

region rendering this fastener the most critical fastener out of all the fasteners tested. However a test of all fasteners around the panel is required to generalize this result.

The strain in wall Type B increases steadily from start to failure with no abrupt change in strain during the entire loading indicating a ductile failure. Wall Type B shows more ductile behavior than wall Type A because of the lack of ability of GWB to deform at higher load in wall Type A where as OSB in wall Type B continues to deform at higher load. Also the OSB panel in wall Type B experiences higher strains than the OSB panel for wall Type A for a given load. In wall Type A, there is higher strain in GWB than in OSB in the initial part of loading. GWB is stiffer than OSB, it attracts load, and in turn deformation is higher than OSB. But being brittle, GWB fails at around 60% of the ultimate wall capacity and load shifts to OSB. This is indicated by a large change in strain in OSB (Figs. 8 and 9). OSB continues to attract load but the strain in OSB increases at a faster rate until failure indicating a much less ductile behavior than that of wall Type B.

Strain Profile

Wall Type A

Each picture in Figs. 10 and 11 is a composite plot showing distribution of local principle strain of areas imaged on different walls using DIC superimposed on a wall image and representing their relative positions on the GWB and OSB side of wall Type A, respectively. Images obtained from seven walls were used to compile the plots, whereas contour plots for the rest of the four

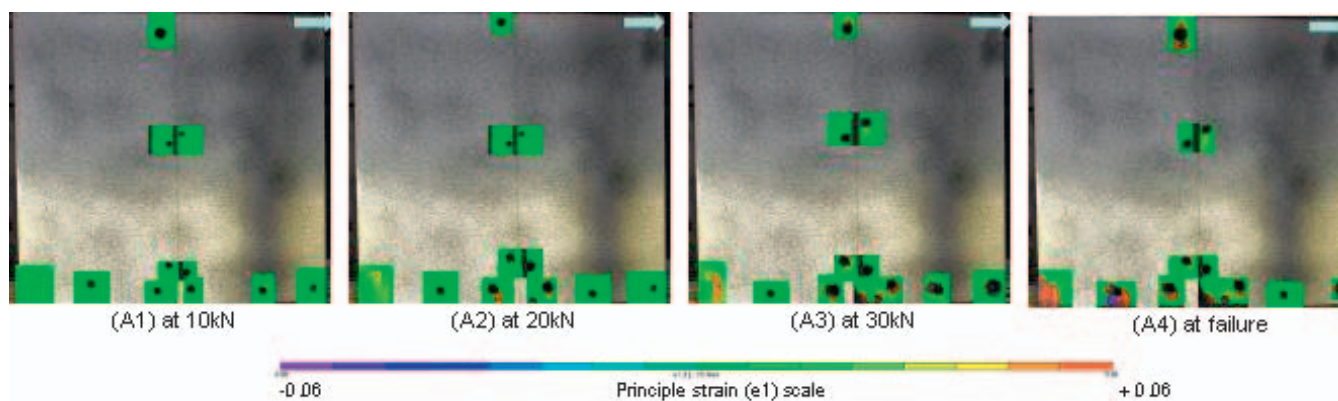


Fig. 10. (Color) Strain profile in GWB at various stages of loading

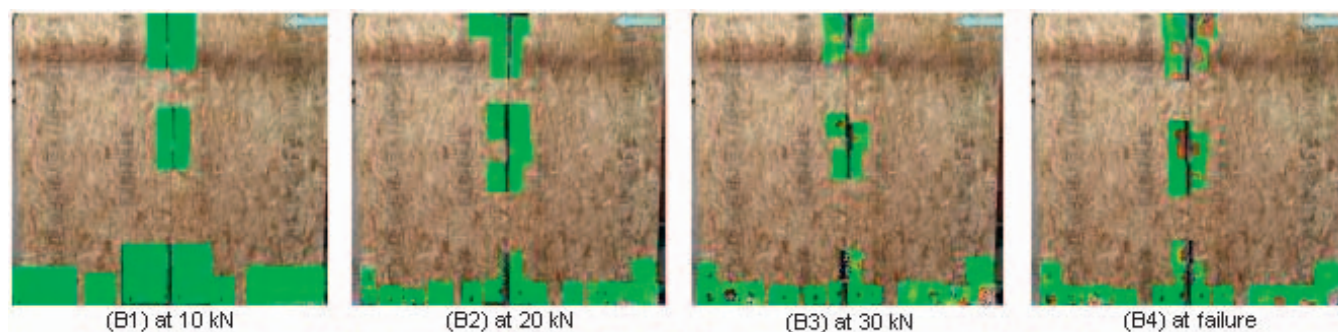


Fig. 11. (Color) Strain profile in OSB (Wall Type A) at various stages of loading

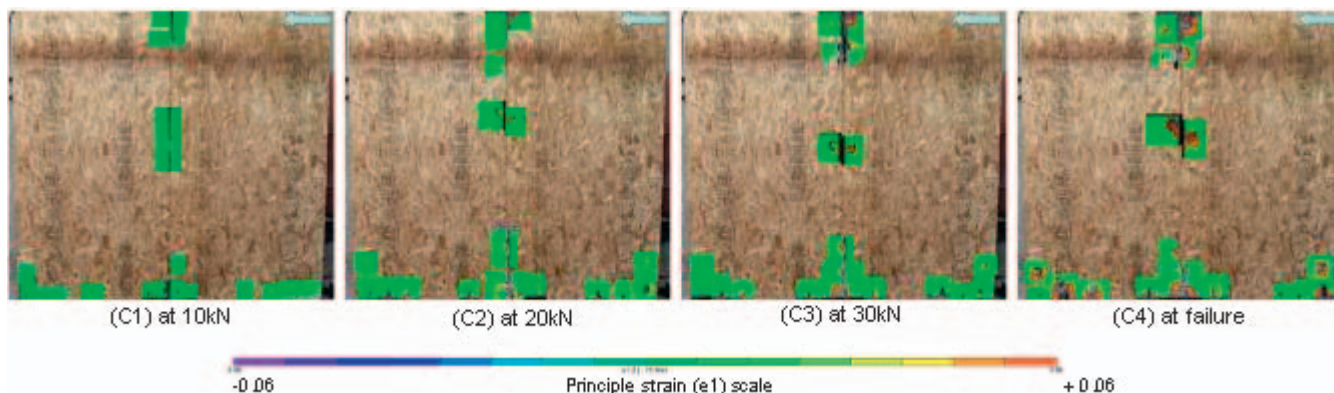


Fig. 12. (Color) Strain profile in OSB (Wall Type B) at various stages of loading

walls can be found in Sinha (2007). The arrow indicates the direction of loading. The loading arrows are reversed because one side (e.g., GWB) is on the back side of the other side (e.g., OSB). All the profiles discussed in this section are the contour profiles of principle strain (ϵ_1) in the material. The numbering scheme for the fasteners is shown in Fig. 5.

Figs. 10 (A1–A4) show the progressive distribution of strain around fasteners for the GWB up to failure. The dark green color shows that the strain in this area is below the detection limit of the system. All the various color contours show the tensile and compressive strain in accordance with contour scale shown in Fig. 10. Similar profiles for the OSB side are shown in Figs. 11 (B1–B4) which has the same scale of reference as Fig. 10. Shades of green indicate no strain while red and purple indicate compressive and tensile strains, respectively. As seen in both figures, there is no strain in the field of the panel and all the strain is concentrated around the fasteners on both sides of the wall.

At 10 kN (Figs. 10 and 11) there is hardly any noticeable strain in either GWB or OSB. Most of the panel is colored in shades of green, hence there is almost no detectable strain in that area. As load is increased from 10 to 20 kN the OSB side [Fig. 11 (B2)] does not experience any strain except Fastener 9 in Zone II, around which a slight increase of strain is observed. However, on the GWB side, the strain starts to appear near Fasteners S8, S9, and S6 [Fig. 10 (A2)]. The onset of strain near the fasteners in GWB implies some load is being transferred through GWB at this initial stage of loading. As GWB is stiffer than OSB, it attracts load and hence higher strains are observed in the GWB panel.

As the load increases to 30 kN, the connections at S9 (Zone II) and S6 (Zone III) have already failed because numerically high and erratic values of strain are calculated by the optical DIC system. Also the panel corner near S6 is significantly deformed [Fig. 10 (A3)]. Strain near other fasteners such as S8, S7, S11, and S13 are also building up. Strains in the uplift corner (Zone II) on the OSB side are clearly visible and are highly concentrated over Fastener 9 [Fig. 11 (B3)]. Considerable buildup of strain in the panel localized to the fastener can be observed along the panel edge at the middle part of the wall while no strain is observed in the field of the panel. At failure, high strain concentrations around most of the fasteners are observed in GWB [Fig. 10 (A4)].

Wall Type B

Fig. 12 shows the strain profile in OSB for wall Type B. Data from four walls were used to generate the plot. The plot is generated in a manner similar to that of Figs. 10 and 11 and uses the

same scale of reference as shown in Fig. 12. At 10 kN [Fig. 12 (C1)] there is hardly any strain in the OSB panel as everything is green. As the load increases to 20 kN, strain has started to build up near Fasteners 4, 5, 8, 9, and 13.

Up to 30 kN there is a steady strain buildup in the Type B walls [Fig. 12 (C3)] but all localized to the fastener and in the field of the panel the strains are below the detectable range. Nails 5 and 13 have a considerable amount of strain and are at the verge of failure as optically recorded deformations are high. Strain is being concentrated around the fasteners which are at the joint of the two panels, making Zones III and IV the critical zones for wall Type B. At failure most of the nails have strain around them signifying failure of the connections while there is no strain in the field of the panel.

Failure Progression

GWB

Strains are observed around Fasteners S9, S8, and S6 at an early stage of loading [Fig. 10 (A2)]. But in terms of magnitude much higher strains are generated near Fastener S9 (1%) as compared to S8 (0.1%) and S6 (0.2%). At 30 kN [Figure 10 (A3)], the area around S9 has already failed. A look at the load-strain curve for that region (Fig. 7) indicates that S9 started failing around 23 kN, i.e., around 60% of wall capacity and has completely failed around 28 kN.

At 30 kN, strains around other fasteners near the sill plate are high and as the load is increased, the areas around these fasteners also begin to fail. At the ultimate wall capacity [Fig 10 (A4)], the area around S9 has completely failed whereas there is sign of failure around other fasteners near the sill plate. It is clear from Fig. 10 that failure of the wall is initiated at S9, i.e., in GWB near the uplift corner (Zone II). Therefore S9 in Zone II is the critical fastener on the GWB side

The panel at the uplift corner undergoes enormous deformation at failure of the panel, as high strains are recorded in that region. As the GWB fails and the stress increases, the paper cover of GWB tear opens, causing rupture of the material within it. This leads to excessive deformation of that part of the panel. Most of the strain is recorded either near the sill and the top plate, while the strains in the central portion (Zone IV) of the wall in GWB are lower than the top plate (Zone V) and the sill (Zones II and III). As most of the strain in GWB is around the fasteners at the sill, GWB predominantly transfers load at the sill level.

Table 4. EEEP Curve Parameters

Parameters (average value)	Walls with OSB only (Type B)	Walls with OSB and GWB (Type A)	Percent increase (+) decrease (−) (%)	COV (%)	
				Type A	Type B
P_{peak} (kN)	38.4	38.7	0.8	6.8	2.6
K_e (kN/mm)	1.4	2.1	49.7	13.4	13.9
μ	4.0	4.8	20.3	18.5	18.4
D	3.3	3.7	13.2	16.2	17.5
Δ_{peak} (mm)	75.9	62.5	−17.7	11.2	7.4
Δf_{ailure} (mm)	93.4	81.5	−12.7	12.3	7.2
Δ_{yield} (mm)	23.8	17.3	−27.3	17.3	14.0
Energy dissipated (J)	2742.0	2466.0	−10.1	18.3	11.7

OSB

During the initial part of loading from 0 to 10 kN OSB both Type A and B walls have low strains in the panels [Fig. 11 (B1) and 12 (C1)]. As the load increases and reaches 20 kN, Type A walls still have strains in the undetectable range [Fig. 11 (B2)] while Type B walls [Fig. 12 (C2)] start to experience some strain around fasteners. Localized strain fields can be observed around Fasteners 4, 5, 8, 9, and 13 in wall Type B. At 30 kN steady buildup of strain is observed in wall Type B. In Type A walls [Fig. 11 (B3)] more buildup of strain is observed around the fastener in the uplift corner (Zone III) and over the joint of the panels (Zones IV and V), while in Type B walls Nails 5 and 13 are on the verge of failure as the deformations recorded optically are high, and strain is being concentrated all around the fasteners which are at the joint of the two panels (Zones III, IV, and V). At failure, in Fig. 12 (C4), strains around all the fasteners are observed, while for Type A walls [Fig. 11 (B4)] no strain concentrations are observed near some fasteners in Zones I, III, and VI. However, fasteners along the middle of the wall (Type A), where the two panel edges meet and the fasteners in the uplift region; have high strain concentrations around them at failure while not much of the strain is concentrated near the fasteners at the sill plate. While OSB transfers load all around but more strain is observed in the middle of the wall where the long edges of two OSB panels meet. As the middle post comprises a single stud, there is inadequate edge distance for the fasteners in that region hence decreasing the connection stiffness in that region and as a result more strain is observed.

At failure most fasteners exhibit high strains around them, but the failure progression is different in Type A and B walls. For OSB in Type A walls, the failure starts at the uplift corner near Nail 9 at approximately 30 kN [Fig. 11 (B3)], followed by the nails in the middle of the walls (11–12) where the panels meet, and then fails in Zone V. The other fasteners, such as 2, 4, 6, and 7, all show some strain around them at failure but not as much as Nails 9, 11, and 12. As onset of strain at the uplift corner nails is earlier than the other nails, making the uplift corner a critical zone for the shear wall assembly. However more tests are needed to confirm this as not all the fasteners were imaged in this study. As more fasteners are imaged in future tests, onset of strain around some other fastener is possible making that the critical fastener. This phenomenon can be due to the fact that gypsum fails at that corner first and the load shifts to the OSB panel. Also these walls are fully anchored, hence sheathing transfers overturning forces into the wall end studs, and subsequently into the foundation through the hold downs, which makes the fasteners in the vicinity of hold down critical.

As in the case of OSB panels in Type A walls, strain near Fastener 9 in Type B walls also starts to increase in the initial stage of loading, and then as it is loaded further, fails at around 23 kN (Fig. 8), and hence is the initiation of failure. For Type B walls, the critical zone is the central region of the wall, where the edges of two panels meet as failure occurs there next. Hence for both types of wall Nail 9 is of prime importance as the failure is initiated from that region. Also, nails in Zones III, IV, and V, which are in the middle stud of the wall, experience a large number of strains.

Comparing the failure pattern of OSB in wall Types A and B, it is observed that wall Type A mostly fails in the middle and some in the bottom near the sill, whereas wall Type B fails near the sill and also in the middle of the panels. Analyzing failure progression in Type A walls, the shear wall fails first at the uplift corner (Zone II) of the gypsum side. Failure continues on to the uplift corner (Zone II) of the OSB side. Finally Zone V of the OSB side fails, leading to failure of the wall. The failure in wall Type B is uniform over all the fasteners imaged, which is the preferred kind of failure as all the fasteners contribute toward the transfer of load to foundation and indicate more efficient design. The presence of GWB in wall Type A prevents the OSB in wall Type A from failing in a manner similar to that of wall Type B. GWB is stiffer than OSB; it attracts load and in turn deformation is higher than OSB. But because GWB is brittle it fails at around 60% of the ultimate wall capacity predominantly near the sill plate signifying that it carries more load in that area until its failure than OSB, and then the load shifts to OSB, hence sharing the load with OSB, and in turn preventing failure of OSB near the sill plate.

Different load paths ensure different failure progression for Type A and B walls. For wall Type A, failure is initiated in GWB near Fastener S9 (Zone II). As the test progresses Fastener 9 on OSB sides subsequently fails and then Zone V in GWB and OSB fails. For wall Type B, failure is initiated in the uplift corner and Fastener 9 goes to Zone III. At failure although all zones show high strain in them but it is Zones IV and V that exhibit more damage.

Contribution of GWB

A summary of results from all the 16 monotonic tests are presented in Table 4. Typical load-deflection curves for both wall types are shown in Fig. 13. As shown in Table 4 and Fig. 13, the contribution of GWB toward the strength of the wall is marginal (0.8%), whereas elastic shear stiffness increased by 50%. As GWB is stiffer than OSB, it contributes to the overall stiffness of

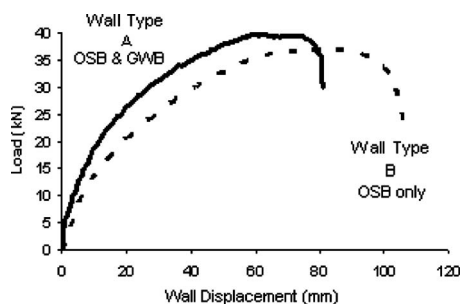


Fig. 13. Typical load-deflection curve for Types A & B walls

the wall but not toward the strength. As shown in Table 4 GWB in the shear wall system reduces the yield, peak, and failure displacements (deformation capacity) by 27, 18, and 13%, respectively. As a result the walls sheathed only with OSB dissipate more energy on its way to failure than a wall with both OSB and GWB. This is observed through Fig. 13 as the wall Type B curve has greater area under it than wall Type A. This is because GWB is brittle, fails early, and does not provide any resistance after its failure and restricts the overall displacement of the wall, whereas OSB continues to provide resistance and deflects a lot more before completely failing. Uang and Gatto (2003) found a 12% increase in shear wall strength, 60% increase in initial stiffness of the wall, and a decrease in deformation capacity of 31% by adding GWB. The differences are probably due to loading conditions and types of fasteners used.

Toothman (2003) and Karacabeyli and Ceccotti (1996) also found an increase in strength of walls when GWB is added. Toothman also found a decrease in deformation capacity of the wall by 10% which is similar to the current study. Toothman concluded that energy dissipated by both types of walls was approximately equal. The results of this study are different from Toothman (2003) and Karacabeyli and Ceccotti (1996) due to variation in the size of walls and different fasteners used for attaching OSB and GWB to the frame. This study uses ring shank nails and standard dry wall screws as opposed to smooth shank nails, for both OSB and GWB, used by Toothman (2003) and Karacabeyli and Ceccotti (1996). Ring shank nails have greater withdrawal values so a higher amount of energy is needed for failure.

As shown in Table 4, the yield displacement decreases on addition of GWB by an average value of 6.5 mm. Also the peak displacement and failure displacement decrease by an average of 13.4 and 12 mm, respectively, on addition of GWB in shear wall assembly. When GWB is included the ductility factor increased by a substantial amount (20%) and ductility is increased by 13%. This can be attributed to the increase in elastic stiffness, which decreased yield displacement. Ductility values alone do not provide much insight into the performance of the walls and is a function of elastic stiffness, yield displacement, and failure displacement. Elastic stiffness can vary with the amount of initial load, which affects the yield point and in turn ductility. As ductility factor is a ratio of failure displacement to yield displacement, a small decrease in yield displacement tends to have a major effect on ductility values. In this study the yield displacement decreases by 27%, which in turn increases the ductility parameters. Also seen in Table 4, ductility of wall Type A is greater than that of Type B while the energy dissipated to failure is less. This is due to a standard way of calculation of ductility parameters which is dependent on yield displacement, failure displacement,

and peak displacements, which in turn also have standard methods for calculation. Numerical parameters should always be seen in conjunction with graphical tools available to determine the true characteristics of the walls; hence all the parameters should be seen in conjunction and not in isolation.

For wall Types A and B the load path is different as the presence of GWB in the assembly alters the way load is carried by the shear wall system. For the sake of redundancy in the system wall Type A, with dual load paths, one through GWB and another through OSB are preferred. Although shear wall is designed assuming that transfer of load is through OSB only, the presence of GWB ensures redundancy, but as GWB transfers the bulk of the loading initially, it defeats the purpose of the design.

We can infer from Figs. 8 and 9 the fact that there is a steady buildup of strain in the OSB panel of Type B walls until failure. The onset of strains is delayed when GWB is present in shear wall assembly. Hence after the onset of strain, deformation of the panel is high in a short duration of time while a steady buildup in wall Type B ensures more ductile behavior of the panel. The magnitude of strain is also greater for wall Type B than that of Type A walls for a given load.

For the assumption in design that OSB carries all the load during a seismic event to be true, the load-strain curve for corresponding areas for OSB in wall Types A and B has to be similar and should have the same generic shape. The differences in the two curves indicate an aspect of load sharing between structural sheathing and GWB which is not incorporated in the design process.

The failures in the OSB panels were all around the fasteners and no strains were observed in the field. The high concentration of strain around the fastener in OSB indicates stress concentration near the fasteners while there is none in the field of the panel. Hence the whole panel is not being utilized for the purpose of transferring shear and only the area in the vicinity of fasteners is being used. The stress needs to be carried by the whole panel to justify an efficient design and this could be done by designing a panel that addresses this issue, or changing the nailing pattern that ensures adequate strength is developed in the wall and also that a majority of the panel area is being utilized in carrying the load.

Although GWB does not increase the load carrying capacity of the wall and it does reduce the capacity to dissipate energy and alters the way OSB carries load, it cannot be done away with in practice because of aesthetics and fire rating of the structure. However arrangements can be made to ensure that the load is transferred only through OSB and GWB is structurally isolated. This can be achieved by designing some innovative connections which will not rigidly attach GWB to the wooden frame behind and allowing it to move as the wall moves without deforming. Another option might be to sheath both sides with OSB and then on one side attach the GWB on the top of OSB but not attach it to the frame structurally so that stiffness of either side of wall is approximately the same and hence ensure equal sharing of load. Further research in this field is required to develop a more accurate and efficient design procedure.

Conclusions

Conclusions based on results of this study include the following:

1. Load is shared by both OSB and GWB initially in a shear wall assembly. GWB fails at around 60% of the actual wall strength and once GWB fails load shifts to OSB;
2. There is a steady buildup of strain in wall Type B from start

to failure and there is no abrupt change in strain during entire loading indicating a ductile failure. Wall Type B shows more ductile behavior than wall Type A because of the lack of ability of the GWB to deform at higher load in wall Type A whereas OSB in wall Type B continues to deform at higher load;

3. Higher strains are observed in GWB during the initial part of loading. GWB is stiffer than OSB, hence attracts more load and in turn deformation is higher than OSB. OSB in walls with GWB (Type A) experiences lower strains than the walls with OSB only (Type B) throughout the loading. The strain in OSB in Wall A increases at a higher rate after the failure of GWB;
4. Strains in OSB and GWB are both concentrated around the fasteners. Strains in the field of the panel were below the detection limit;
5. The load path for both wall types is different. Failure in wall Type A starts at the uplift corner in GWB and then moves to the uplift corner in OSB. Finally the walls fail at the middle of the top plate for both GWB and OSB. In wall Type B the failure is initiated at the uplift corner in OSB followed by the middle region at the sill level and ends up at the middle section of the wall where two panels meet. The uplift corner fasteners are of prime importance in both types of walls and panels;
6. GWB does not contribute toward the overall strength of the shear wall, but it increases the stiffness of the wall by 50%. GWB is stiffer than OSB, and hence considerably contributes to stiffness; and
7. Ductility factor (μ) of the system increases by 20% and the ductility of the system increases by 13% while energy dissipated by the wall decreases when GWB is included in the shear wall assembly. GWB being brittle reduces the ability to deform before failing and hence 18, 13, and 27% decrease is observed in peak, failure, and yield displacements, respectively.

Recommendations based on the results of this study include the following:

1. Further tests and supporting results are needed to generalize the failure progression pattern for both types of walls;
2. All the walls had ringed shank nails that eliminate withdrawal as a mode of failure. Further testing is required with conventional nails for a conclusion that could be generalized;
3. Cyclic and dynamic tests of walls using DIC should be done to provide a complete picture of shear wall behavior during a seismic event;
4. Study should be conducted on all the other fasteners which are omitted in this study. Study on other fasteners could reveal more about failure progression and identify new critical zones;
5. The effect of GWB in shear wall system needs to be considered, either by incorporating it in the design for damage limitation or by structurally detaching GWB from the shear wall frame by means of innovative connections;
6. The aspect of load sharing needs to be quantified to develop efficient design procedure; and
7. More efficient connection or panel or both designs are needed to utilize the whole panel for shear transfer.

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